

1. (13 points) This problem concerns a Normal Bayesian regression model that predicts the time to failure of microchips (in hours) as a function of their solder temperature (in degrees Celsius). The model is fit using default priors in the `stan_glm` function.

- (a) Figures 1-3 on the supplemental sheet are trace plots, density plots, and autocorrelation plots of the MCMC simulation. Describe what each plot tells us about whether or not the MCMC simulation has converged to a good posterior approximation.

Solution: Figure 1 shows us that the chain has mixed well for each of the variables. Figure 2 shows us that each chain has produced approximately the same posterior distribution for each variable. Figure 3 shows us that we don't have an issue with autocorrelation and thus the chain is producing values that act like they are independent. Overall the simulation has converged well.

- (b) Draw an example of a trace plot that would represent a slow mixing chain.

Solution: Answers will vary.

- (c) (Bonus) One of the metrics we calculate for MCMC diagnostics is R-hat. Explain what R-hat measures.

Solution: R-hat measures the ratio of the combined and within root variability. Thus a value of 1 means that the chain is stable, and values greater than 1 represent instability.

- (d) Suppose the R-hat value from this MCMC is 1.4. Would this be a cause for concern based on our rule of thumb from class? (No explanation necessary).

Solution: Yes

- (e) Consider the following output from the `prediction_summary` function for this model. What does the value of 0.5909091 in the output indicate?

mae	mae_scaled	within_50	within_95
638.343	0.5191722	0.5909091	1

Solution: About 59 percent of the original values in the data set fall within the 50% posterior predictive interval calculated for the model.

- (f) Figure 4 on the supplemental sheet plots the densities of 50 posterior predictive distributions versus the density of the original data. Explain what this plot says about the assumptions of our regression model.

Solution:

2. (7 points) Consider a data set that contains ratings from the website *ratemyprofessors.com* about professors at a large Midwestern university compiled by Bleske-Rechek and Fritsch in 2011. The response variable is *attractive*, which is 1 if the consensus among the raters is that the professor is attractive, and 0 otherwise. The predictors are the *gender* of the professor (male or female) and the average *easiness* rating of the professor on a scale of 1 to 5, with 5 being the easiest. See the supplemental sheet for output of a Bayesian logistic regression model named *profmod*.

- (a) Using the median posterior parameter values from the output, estimate the probability that a female professor who has an average easiness rating of 4.2 will be considered attractive.

Solution: $-5.49 + 0.179 + 1.03 \cdot 4.2 = -0.985$
 $\exp(-0.985) / (1 + \exp(-0.985)) = 0.2719008$
So there is about a 27% chance this professor is considered attractive.

- (b) Suppose a male and a female professor both have a easiness rating of 3.1. The odds that the female professor is considered attractive is estimated to be how many times greater than the odds the male professor is considered attractive?

Solution: $\exp(.179) = 1.19$

- (c) Give an interpretation of the exponentiated interval for the easiness variable (the one that goes from 1.94 to 4.14).

Solution: There is a 90% posterior probability that for every additional point in a professor's average easiness rating, the odds that they are considered attractive increases by a multiplicative factor of between 1.94 and 4.14 for professors of the same gender.